

# Scanner Development Documentation

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## 1. Get barcode

**Default Action: android.intent.ACTION\_DECODE\_DATA**

**Default data receiving KEY: barcode\_string**

Note: Through code calling, the output mode we usually use is: broadcast mode. The broadcast mode requires setting the PDA output mode to broadcast mode through the SDK API. The specific operations are as follows:

### 1.1 Set the output mode to [Broadcast Mode]

```
ScanManager mScanManager = new ScanManager();  
mScanManager.switchOutputMode(0); //0: Broadcast mode 1: Keyboard mode
```

### 1.2 Register Broadcast

```
IntentFilter filter = new IntentFilter();  
filter.addAction(ScanManager.ACTION_DECODE);  
registerReceiver(mReceiver, filter);
```

### 1.3 Listening for broadcast callbacks

```
private BroadcastReceiver mReceiver = new BroadcastReceiver() {  
    @Override  
    public void onReceive(Context context, Intent intent) {  
        String barcodeStr = intent.getStringExtra(ScanManager.BARCODE_STRING_TAG  
    ); //barcodeStr gets the scanned barcode data  
    }  
};
```

### 1.4 Cancel registration (will not receive broadcast results)

```
unregisterReceiver(mReceiver);
```

## 2. interface

### 2.1 switchOutputMode [ Set output mode ]

|                              |                                                                 |             |                                                    |
|------------------------------|-----------------------------------------------------------------|-------------|----------------------------------------------------|
| <b>definition</b>            | public boolean switchOutputMode(int mode) ;                     |             |                                                    |
| <b>illustrate</b>            | Set the barcode output method after the PDA scans the barcode . |             |                                                    |
| <b>parameter</b>             | <b>name</b>                                                     | <b>type</b> | <b>Remark</b>                                      |
|                              | mode                                                            | int         | 0: Broadcast mode (Intent output) 1: Keyboard mode |
| <b>return</b><br>( boolean ) | true : Setting successful<br>false: Setting failed              |             |                                                    |
| <b>Reference code</b>        | mScanManager.switchOutputMode(0);                               |             |                                                    |

### 2.2 getOutputMode [ Get output mode ]

|                          |                                                           |             |               |
|--------------------------|-----------------------------------------------------------|-------------|---------------|
| <b>definition</b>        | public int getOutputMode ();                              |             |               |
| <b>illustrate</b>        | Get the scan result output mode currently used by the PDA |             |               |
| <b>parameter</b>         | <b>name</b>                                               | <b>type</b> | <b>Remark</b> |
|                          |                                                           |             |               |
| <b>return</b><br>( int ) | 0: Broadcast mode (Intent output)<br>1: Keyboard mode     |             |               |
| <b>Reference code</b>    | mScanManager.getOutputMode();                             |             |               |

### 2.3 startDecode 【 Start scanning 】

|                   |                                                                                                                                                                                     |             |               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b> | public boolean startDecode() ;                                                                                                                                                      |             |               |
| <b>illustrate</b> | The code starts the scanning head to scan the code (same effect as the left and right scanning buttons. If you use buttons to trigger scanning, you don't need to call this method) |             |               |
| <b>parameter</b>  | <b>name</b>                                                                                                                                                                         | <b>type</b> | <b>Remark</b> |

|                              |                                                    |  |  |
|------------------------------|----------------------------------------------------|--|--|
|                              |                                                    |  |  |
| <b>return</b><br>( boolean ) | true : Setting successful<br>false: Setting failed |  |  |
| <b>Reference code</b>        | mScanManager.startDecode();                        |  |  |

## 2.4 stopDecode 【 Stop scanning 】

|                              |                                                    |             |               |
|------------------------------|----------------------------------------------------|-------------|---------------|
| <b>definition</b>            | public boolean stopDecode () ;                     |             |               |
| <b>illustrate</b>            | Code control stops scanning.                       |             |               |
| <b>parameter</b>             | <b>name</b>                                        | <b>type</b> | <b>Remark</b> |
|                              |                                                    |             |               |
| <b>return</b><br>( boolean ) | true : Setting successful<br>false: Setting failed |             |               |
| <b>Reference code</b>        | mScanManager.stopDecode();                         |             |               |

## 2.5 setTriggerMode 【 Set trigger scan mode 】

|                              |                                                    |             |                                                                                                                                                       |
|------------------------------|----------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>definition</b>            | public void setTriggerMode(TriggeringtgMode);      |             |                                                                                                                                                       |
| <b>illustrate</b>            |                                                    |             |                                                                                                                                                       |
| <b>parameter</b>             | <b>name</b>                                        | <b>type</b> | <b>Remark</b>                                                                                                                                         |
|                              | tgMode                                             | Triggering  | Triggering.PULSE Short press triggers decoding or timeout stops<br>Triggering. CONTINUOUS mode<br>Triggering.HOST long press trigger, release to stop |
| <b>return</b><br>( boolean ) | true : Setting successful<br>false: Setting failed |             |                                                                                                                                                       |
| <b>Reference code</b>        | mScanManager.setTriggerMode(Triggering.PULSE);     |             |                                                                                                                                                       |

## 2.6      `getTriggerMode` [ Get trigger scan mode ]

|                                                   |                                                                                                                                                                                                  |             |               |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b>                                 | <code>public Triggering getTriggerMode ();</code>                                                                                                                                                |             |               |
| <b>illustrate</b>                                 |                                                                                                                                                                                                  |             |               |
| <b>parameter</b>                                  | <b>name</b>                                                                                                                                                                                      | <b>type</b> | <b>Remark</b> |
|                                                   |                                                                                                                                                                                                  |             |               |
| <b>return</b><br>( <b>Triggerin</b><br><b>g</b> ) | <b>Triggering.PULSE</b> Short press triggers decoding or timeout stops<br><b>Triggering.CONTINUOUS</b> Continuous scanning mode<br><b>Triggering.HOST</b> Long press to trigger, release to stop |             |               |
| <b>Reference</b><br><b>code</b>                   | <code>mScanManager.getTriggerMode();</code>                                                                                                                                                      |             |               |

## 2.7      `lockTriggler` [ Disable scanning ]

|                                     |                                                                          |             |               |
|-------------------------------------|--------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b>                   | <code>public boolean lockTrigger() ;</code>                              |             |               |
| <b>illustrate</b>                   | Set the scan trigger inactive (disable the scan button and scan method). |             |               |
| <b>parameter</b>                    | <b>name</b>                                                              | <b>type</b> | <b>Remark</b> |
|                                     |                                                                          |             |               |
| <b>return</b><br>( <b>boolean</b> ) | <b>true</b> : Setting successful<br><b>false</b> : Setting failed        |             |               |
| <b>Reference</b><br><b>code</b>     | <code>mScanManager.lockTriggler();</code>                                |             |               |

## 2.8      `unlockTriggler` [ Enable scanning ]

|                   |                                               |
|-------------------|-----------------------------------------------|
| <b>definition</b> | <code>public boolean unlockTrigger() ;</code> |
|-------------------|-----------------------------------------------|

|                                     |                                                                   |             |               |
|-------------------------------------|-------------------------------------------------------------------|-------------|---------------|
| <b>illustrate</b>                   | <b>Set the scan trigger to activate (enable the scan button).</b> |             |               |
| <b>parameter</b>                    | <b>name</b>                                                       | <b>type</b> | <b>Remark</b> |
|                                     |                                                                   |             |               |
| <b>return</b><br><b>( boolean )</b> | <b>true : Setting successful</b><br><b>false: Setting failed</b>  |             |               |
| <b>Reference code</b>               | <b>mScanManager.unlockTrigger();</b>                              |             |               |

## 2.9 openScanner [ Open the scanning engine ]

|                                     |                                                                                                               |             |               |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b>                   | <b>public boolean openScanner ();</b>                                                                         |             |               |
| <b>illustrate</b>                   | <b>Open the scanning engine (usually it is turned on by default, so you don' t need to call this method).</b> |             |               |
| <b>parameter</b>                    | <b>name</b>                                                                                                   | <b>type</b> | <b>Remark</b> |
|                                     |                                                                                                               |             |               |
| <b>return</b><br><b>( boolean )</b> | <b>true : Setting successful</b><br><b>false: Setting failed</b>                                              |             |               |
| <b>Reference code</b>               | <b>mScanManager.openScanner();</b>                                                                            |             |               |

## 2.10 closeScanner 【 Close the scanning engine 】

|                                     |                                                                                         |             |               |
|-------------------------------------|-----------------------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b>                   | <b>public boolean closeScanner() ;</b>                                                  |             |               |
| <b>illustrate</b>                   | <b>Disable the scanning engine. Scanning will not be possible after it is disabled.</b> |             |               |
| <b>parameter</b>                    | <b>name</b>                                                                             | <b>type</b> | <b>Remark</b> |
|                                     |                                                                                         |             |               |
| <b>return</b><br><b>( boolean )</b> | <b>true : Setting successful</b><br><b>false: Setting failed</b>                        |             |               |



|                           |                                           |
|---------------------------|-------------------------------------------|
| <b>Reference<br/>code</b> | <code>mScanManager.closeScanner();</code> |
|---------------------------|-------------------------------------------|

## 2.11 `setParameterString` 【 Set scan parameters (value is String type) 】

|                               |                                                                                                                                                                                                                                               |             |                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>definition</b>             | <code>public boolean setParameterString(int[] index, String[] value) ;</code>                                                                                                                                                                 |             |                                                                                                                                       |
| <b>illustrate</b>             | Setting Scan Parameters                                                                                                                                                                                                                       |             |                                                                                                                                       |
| <b>parameter</b>              | <b>name</b>                                                                                                                                                                                                                                   | <b>type</b> | <b>Remark</b>                                                                                                                         |
|                               | index                                                                                                                                                                                                                                         | int[]       | Set the parameter key corresponding to the parameter. The length becomes longer and a parameter array of any length can be passed in. |
|                               | value                                                                                                                                                                                                                                         | String[]    | Set the corresponding value of the corresponding parameter. The length should correspond to the length of the key above.              |
| <b>return<br/>( boolean )</b> | true : Setting successful<br>false: Setting failed                                                                                                                                                                                            |             |                                                                                                                                       |
| <b>Reference<br/>code</b>     | <code>mScanManager..setParameterString(new<br/>int[]{PropertyID.WEDGE_INTENT_ACTION_NAME,PropertyID.WEDG<br/>E_INTENT_DATA_STRING_TAG},new<br/>String[]{"com.pkuhit.intent.action.SCANRECVR","KeyString           you<br/>received"});</code> |             |                                                                                                                                       |

## 2.12 `getParameterString` [ Get scan parameters (String type) ]

|                   |                                                                             |             |               |
|-------------------|-----------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b> | <code>public String[] getParameterString(int[] index) ;</code>              |             |               |
| <b>illustrate</b> | Get scan parameters                                                         |             |               |
| <b>parameter</b>  | <b>name</b>                                                                 | <b>type</b> | <b>Remark</b> |
|                   |                                                                             |             |               |
| <b>return</b>     | The subscript of the String[] result corresponds to the corresponding value |             |               |

|                   |                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ( String[] )      | of the subscript parameter of the passed parameter.                                                                                                                      |
| Reference<br>code | <pre> int[] keys = new int[]{PropertyID.WEDGE_INTENT_ACTION_NAME,PropertyID.WEDG E_INTENT_DATA_STRING_TAG}; String[] ret = mScanManager.getParameterString(keys); </pre> |

### 2.13 setPropertyInts [Set scanning parameters (value is int type)]

|                      |                                                                                                                                                                                                         |       |                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| definition           | public boolean setPropertyInts(int[] index, int[] value);                                                                                                                                               |       |                                                                                                                                       |
| illustrate           | Setting Scan Parameters                                                                                                                                                                                 |       |                                                                                                                                       |
| parameter            | name                                                                                                                                                                                                    | type  | Remark                                                                                                                                |
|                      | index                                                                                                                                                                                                   | int[] | Set the parameter key corresponding to the parameter. The length becomes longer and a parameter array of any length can be passed in. |
|                      | value                                                                                                                                                                                                   | int[] | Set the corresponding value of the corresponding parameter. The length should correspond to the length of the key above.              |
| return<br>(boolean ) | true : Setting successful<br>false: Setting failed                                                                                                                                                      |       |                                                                                                                                       |
| Reference<br>code    | <pre> mScanManager.setPropertyInts(new int[]{PropertyID.SEND_GOOD_READ_BEEP_ENABLE,PropertyID.SEN D_GOOD_READ_VIBRATE_ENABLE},new int[]{2,1}); ////Turn on sound and vibration in broadcast mode </pre> |       |                                                                                                                                       |

### 2.14 getPropertyInts [Get scan parameters (int type)]

|                           |                                                                                                                                                                                                            |             |               |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|
| <b>definition</b>         | <b>public int[] getPropertyInts(int[] index);</b>                                                                                                                                                          |             |               |
| <b>illustrate</b>         | <b>Get scan parameters</b>                                                                                                                                                                                 |             |               |
| <b>parameter</b>          | <b>name</b>                                                                                                                                                                                                | <b>type</b> | <b>Remark</b> |
|                           |                                                                                                                                                                                                            |             |               |
| <b>return<br/>(int[])</b> | <b>The subscript of the int[] result corresponds to the corresponding value of the subscript parameter of the passed parameter.</b>                                                                        |             |               |
| <b>Reference<br/>code</b> | <b>int[] keys = new</b><br><b>int[]{PropertyID.SEND_GOOD_READ_BEEP_ENABLE,PropertyID.SEN</b><br><b>D_GOOD_READ_VIBRATE_ENABLE};</b><br><b>int[] ret</b><br><b>= mScanManager.getParameterString(keys);</b> |             |               |

### 2.15.resetScannerParameters [Reset scanning head related settings]

|                             |                                                        |             |               |
|-----------------------------|--------------------------------------------------------|-------------|---------------|
| <b>definition</b>           | <b>public boolean resetScannerParameters() ;</b>       |             |               |
| <b>illustrate</b>           | Resets scan head related settings to factory defaults. |             |               |
| <b>parameter</b>            | <b>name</b>                                            | <b>type</b> | <b>Remark</b> |
|                             |                                                        |             |               |
| <b>return<br/>(boolean)</b> | true : Setting successful<br>false: Setting failed     |             |               |
| <b>Reference<br/>code</b>   | <b>mScanManager.resetScannerParameters() ;</b>         |             |               |

### 3. Common setting examples

| Reference table of some property settings in different modes |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model                                                        | Function                                                                       | Reference code                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Broadcast Mode                                               | Turn sound and vibration on or off                                             | <pre> ScanManager mScanManager = new ScanManager(); //The following settings are for the output mode of the scanning head as intent, so there is SEND in the setting property identifier mScanManager.setPropertyInts(new int[] {PropertyID.SEND_GOOD_READ_BEEP_ENABLE},new int[] {0}); //Sound 0 off 1 short 2 sharp //Vibrate when scanning is successful mScanManager.setPropertyInts(new int[] {PropertyID.SEND_GOOD_READ_VIBRATE_ENABLE},new int[] {0}); //Vibration 0 off 1 on </pre> |
|                                                              | Set the action for broadcast mode and the keyString for receiving scan results | <pre> new ScanManager().setParameterString(new int[] {PropertyID.WEDGE_INTENT_ACTION_NAME,PropertyID.WEDG E_INTENT_DATA_STRING_TAG},new String[] {"Your Action","KeyString you received"}); </pre>                                                                                                                                                                                                                                                                                          |
| Keyboard Mode                                                | Turn keyboard mode sounds and vibrations on or off                             | <pre> ScanManager mScanManager = new ScanManager(); //The following settings are for the output mode of the scanning head to be keyboard mode, so there is no SEND in the setting property mark mScanManager.setPropertyInts( new int[] {PropertyID.GOOD_READ_BEEP_ENABLE},new int[] {0}); //Sound 0 off 1 short 2 sharp //Vibrate when scanning is successful mScanManager.setPropertyInts(new int[] {PropertyID.GOOD_READ_VIBRATE_ENABLE},new int[] {0}); //Vibration 0 off 1 on </pre>   |
|                                                              | Keyboard type                                                                  | <pre> //Output mode is keyboard mode new ScanManager().setParameterInts( new int[] {PropertyID.WEDGE_KEYBOARD_TYPE},new int[] {1}); 0: Physical keyboard 1: Input method software 2: Use only physical keyboard </pre>                                                                                                                                                                                                                                                                      |

|  |                                        |                                                                                                                                                                                                                             |
|--|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |                                        | 3: Use input method only                                                                                                                                                                                                    |
|  | <b>Set the operation key character</b> | <pre>new ScanManager().setParameterInts( new int[] {PropertyID.LABEL_APPEND_ENTER},new int[] {1});</pre> //Optional value corresponding to the parameter<br>0 : None<br>1: Carriage return<br>2: IME action done<br>3 : TAB |

| Common function reference table                                             |                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Function                                                                    | Reference code                                                                                                                                                                                                                                                                                                                                                       |
| <b>Set the output mode</b>                                                  | <pre>new ScanManager().switchOutputMode(0); //Turn on broadcast mode new ScanManager().switchOutputMode(1); //Turn on keyboard mode</pre>                                                                                                                                                                                                                            |
| <b>Turn app identifiers on or off</b>                                       | <pre>int [] keyInt = new int[] {PropertyID.LABEL_SEPARATOR_ENABLE}; int [] valueInt = new int [] {1}; //0: off 1: on new ScanManager().setPropertyInts(keyInt,valueInt);</pre>                                                                                                                                                                                       |
| <b>Scan the code to add prefix and suffix, the additional format is 0-6</b> | <pre>new ScanManager(). setPropertyInts(new int[] {PropertyID.SEND_LABEL_PREFIX_SUFFIX},new int[] {1});</pre>                                                                                                                                                                                                                                                        |
| <b>replace</b>                                                              | <pre>new ScanManager() .setParameterString(new int[] {PropertyID.LABEL_MATCHER_TARGETREGEX,PropertyID.LABEL_MATCHER_REPLACE},new String[] {"oldStr","newStr"});</pre>                                                                                                                                                                                                |
| <b>Multiple barcode settings (not necessarily effective)</b>                | <pre>int[] multiConfigbuf = new int[] { PropertyID.MULTI_DECODE_MODE, PropertyID.FULL_READ_MODE, PropertyID.BAR_CODES_TO_READ}; int[] multiConfigval = new int[3]; multiConfigval[0] = 1; //0 turns off the multi-barcode mode, 1 turns on the multi-barcode mode multiConfigval[1] = 1; //0 Generate a decode event after one or more barcodes are decoded 1:</pre> |

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| for every device)                                             | <p>A callback is generated to the application only when at least the set number of barcodes are decoded.</p> <p>multiConfigval[2] = 4; //This parameter sets the number of decoded barcodes, 1-10. The default value is 1.</p> <p>new ScanManager().setParameterInts(multiConfigbuf, multiConfigval);</p> <p>Get:</p> <p>//To obtain multiple barcode scanning results through broadcast, you need to set the output mode of the scanning head to broadcast output.</p> <p>String moreBarcodeStr=intent.getStringExtra("com.ubx.datawedge.data_string");</p> |
| Deduplication settings in continuous scanning mode            | <p>new ScanManager().setParameterInts(<br/> new int[]{PropertyID.DEC_Multiple_Decode_MODE},new int[]{0}); //0: Repeatable decoding 1: Continuous non-repeated decoding 2: Non-repeated decoding in the cache</p>                                                                                                                                                                                                                                                                                                                                             |
| Set the interval time between scanning two identical barcodes | <p>//value * 100 ms; MIN == 1, MAX == 99 Set the interval time between scanning two identical barcodes. For example, if you pass 1, 1*100ms == 100ms; if you pass the maximum value 99, 99*100ms == 9900ms, that is, the interval is 9900 milliseconds</p> <p>mScanManager.setPropertyInts(new<br/> int[]{PropertyID.TIMEOUT_BETWEEN_SAME_SYMBOL},new int[]{1});</p>                                                                                                                                                                                         |
| Set the scan interval and scan timeout                        | <p>ScanManager mScanManager = new ScanManager();</p> <p>mScanManager.setPropertyInts(new<br/> int[]{PropertyID.DEC_Multiple_Decode_INTERVAL},new int[]{50}); //Scan interval, default 50ms (range: 0-5000ms)</p> <p>mScanManager.setParameterInts(<br/> new int[]{PropertyID.DEC_Multiple_Decode_TIMEOUT},new int[]{5000}); //Scan timeout, default 5000ms (range: 50-60000ms)</p>                                                                                                                                                                           |
| Turn the corresponding symbology on or off                    | <p>ScanManager mScanManager = new ScanManager();</p> <p>mScanManager.enableSymbology(Symbology.AZTEC,false);</p> <p>//AZTEC: specific code system, false: do not decode true: decode</p> <p>mScanManager.enableAllSymbologies(true); // Enable decoding of all symbologies, true: Enable false: Disable</p>                                                                                                                                                                                                                                                  |
| Activate aiming mode                                          | <p>mScanManager.setParameterInts(new int[]{PropertyID.DEC_PICKLIST_AIM_MODE},new int[]{1});</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| The default code                                              | <p>int[] codeleveindex = new int[]{ PropertyID.LINEAR_CODE_TYPE_SECURITY_LEVEL};</p> <p>int[] codelevelvalue = new int[]{ 1 }; //Level 0-3.</p> <p>int ret2 = new ScanManager().setParameterInts(codeleveindex, codelevevalue);</p>                                                                                                                                                                                                                                                                                                                          |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| verification level is 1                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Set the symbology and check digit on or off | <pre> int[] codeChecksumKey = new int[]{ PropertyID.EAN13_ENABLE, //Turn on the EAN13 code switch, 0: off 1: on PropertyID.UPCA_ENABLE, //Set UPCA code switch, 0: off 1: on PropertyID.UPCE_ENABLE, //Set UPCE code switch, 0: off 1: on PropertyID.UPCE1_ENABLE, //Set UPCE1 code switch, 0: off 1: on PropertyID.EAN13_SEND_CHECK, //Set UPCE1 check switch, 0: Off 1: On PropertyID.UPCA_SEND_CHECK, //Set UPCA check switch, 0: off 1: on PropertyID.UPCE_SEND_CHECK, //Set the UPCE check switch, 0: Off 1: On PropertyID.UPCE1_SEND_CHECK}; //Set UPCE1 check switch, 0: off 1: on int[] codeChecksumvalue = new int[]{1,1,1,1,0,0,0,0}; int ret = new ScanManager().setParameterInts( codeChecksumKey, codeChecksumvalue); </pre> |
| Low contrast image mode                     | <pre> ScanManager mScanManager = new ScanManager(); mScanManager.setPropertyInts(new int[]{PropertyID.LOW_CONTRAST_IMPROVED,PropertyID.LOW_CONTRAST_IMPRO VED_ALGORITHM},new int[]{2,0}); //Low contrast image mode, value: 0-5 //Enhance low contrast image algorithm 0: Disable 1: Enable </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Light and aiming light settings             | <pre> new ScanManager().setParameterInts( new int[]{PropertyID.DEC_2D_LIGHTS_MODE},new int[]{1}); //1: Aiming light only 2: Lighting light only 3: Alternate operation 4: Simultaneous operation </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Viewfinder                                  | <pre> //DEC_2D_CENTERING_ENABLE value 0 1 Close, open //DEC_2D_CENTERING_MODE framing mode 0, 1, 2, 3 mScanManager.setParameterInts(new int[]{PropertyID.DEC_2D_CENTERING_ENABLE,PropertyID.DEC_2D_CENTERING_M ODE},new int[]{1,2}); //coordinate int[] key = new int[]{ PropertyID.DEC_2D_WINDOW_UPPER_LX, PropertyID.DEC_2D_WINDOW_UPPER_LY,PropertyID.DEC_2D_WINDOW_LOWER_R X,PropertyID.DEC_2D_WINDOW_LOWER_RY }; int[] value = new int[]{632, 346,640,354}; mScanManager.setParameterInts(key, value); </pre>                                                                                                                                                                                                                        |
| Encoding format                             | <pre> int[] key = new int[]{ PropertyID.CODING_FORMAT}; //Chinese encoding format int[] value = new int[]{1}; //0: UTF-8 1: GBK mScanManager.setParameterInts(key, value); </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| OCR                                         | <pre> int[] key = new int[]{ PropertyID.DEC_OCR_MODE,PropertyID.DEC_OCR_TEMPLATE }; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>decoding<br/>mode and<br/>OCR<br/>template</b> | <pre>//DEC_OCR_MODE: mode (0-3) DEC_OCR_TEMPLATE: template (0-4) int[] value = new int[] {3,0}; //3: Both (mode) 0: User Defind (template) int ret = mScanManager.setParameterInts(key, value);</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>OCR<br/>custom<br/>template</b>                | <pre>int[] key = new int[] { PropertyID.DEC_OCR_USER_TEMPLATE }; String[] action = { "1377777777777777770" }; boolean ret = mScanManager.setParameterString(key, action);</pre> <p>String starts with 1 Fixed</p> <p>The second character represents the font (1, 2, 3, 4, 5) 3 represents both ocrA &amp; ocrB fonts</p> <p>The third to the second to last character represents the character type (character represents the character type; a combination of numbers, letters, and spaces, with values of 5 (number combination), 6 (letter combination), 7 (number and letter combination), 8. (Any supported character combination))</p> <p>String ends with 0 Fixed</p> <p>For example, an 18-digit ID card:</p> <p>1377777777777777770</p> |
| <b>Optimize<br/>1D code</b>                       | <pre>ScanManager mScanManager = new ScanManager(); int[] key = new int[] { PropertyID.FUZZY_1D_PROCESSING }; //Optimize one-dimensional code int[] value = new int[] {3 }; //0: Close 1: Open int ret = mScanManager.setParameterInts(key, value);</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Exposure<br/>parameter<br/>settings</b>        | <pre>int[] key = new int[] { PropertyID.IMAGE_EXPOSURE_MODE, PropertyID.DEC_ES_MAX_EXP, PropertyID.DEC_ES_MAX_GAIN, PropertyID.DEC_ES_TARGET_VALUE }; //Exposure parameter settings int[] value = new int[] {1,500,4,4000}; int ret = mScanManager.setParameterInts(key, value);</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



## 4 Get the scan head image (obtained through broadcast mode)

**Note:** Do not use the following two methods at the same time to avoid the problem of not being able to obtain the picture

### 4.1 Active call method

automatically call this method after obtaining the barcode .

#### 4.1.1 Sending Broadcasts

You need to send it once every time before you get the picture.

```
Intent intentImage = new Intent("action.scanner_capture_image");  
//intentImage.putExtra("rotate",180);//Rotate the image angle  
sendBroadcast(intentImage);
```

#### 4.1.2 Receiving image data by broadcasting

Register the broadcast for receiving images :

```
String          action_scanner_capture_image_result          =  
"scanner_capture_image_result";  
  
IntentFilter filter = new IntentFilter();  
filter.addAction(action_scanner_capture_image_result);  
registerReceiver(mScanReceiver, filter);
```

#### 4.1.3 Setting broadcast callback

```
BroadcastReceiver mScanReceiver = new BroadcastReceiver() {  
    @Override  
    public void onReceive(Context context, Intent intent) {
```

```

if(intent != null &&
intent.getAction().equals(action_scanner_capture_image_result)) {
byte[] bmpByte = intent.getByteArrayExtra(" bitmapBytes");
if(bmpByte != null && bmpByte.length > 1) {
Bitmap mBitmap = BitmapFactory.decodeByteArray(bmpByte, 0,
bmpByte.length).copy(Bitmap.Config.RGB_565, true);
}else {
Toast.makeText( ScanActivity.this, "Failed to obtain image" ,
Toast.LENGTH_SHORT).show();
}
}
}
};

```

#### 4.1.4 Cancellation of registration

If you no longer want to receive pictures, you can cancel your registration.

```
unregisterReceiver(mScanReceiver);
```

## 4.2 Automatic return method

**This method is applicable to our company's Android 11 and above devices, and ensure that the OS version is 2024 and later.**

After the program is started, you only need to send a broadcast once. After scanning the code subsequently, the image data will be automatically returned in the broadcast.

### 4.2.1 Sending Broadcast

```
private static String BROADCAST_IMAGE_ACTION =
"com.ubx.scanner_capture_image_result";
```

```
Intent intentImage= new Intent();
intentImage.setAction("com.ubx.barcode.broadcast_image");
intentImage.putExtra("imageAction",BROADCAST_IMAGE_ACTION);
intentImage.putExtra("enableSendImage",true);
intentImage.putExtra(" enableSendFailedImage",true);
intentImage.putExtra("outputImageMode",0);
intentImage.putExtra("jpegQuality", 100);
intentImage.putExtra("outputBounds", true);
intentImage.putExtra("imageFormat",0);
intentImage.putExtra("saveImageDirPath","/storage/emulated/0/imageSa
ve/");
sendBroadcast(intentImage);
```

Code example:

```
/**
 * Set automatic return image
 * @param context
 * @param enableSendImage
 */
private void enableAutoSaveImage(Context context,boolean
enableSendImage){
try {
Intent intentImage= new Intent();
intentImage.setAction("com.ubx.barcode.broadcast_image");

//Set the action for receiving the image result
intentImage.putExtra("imageAction",BROADCAST_IMAGE_ACTION);

//Whether to broadcast the image data and barcode data directly
```

```

after decoding the barcode
intentImage.putExtra("enableSendImage", enableSendImage);

//Whether to send the decoding failure image by default (some
scanner terminals do not support this, please consult our technical
support for details)
intentImage.putExtra("enableSendFailedImage", enableSendImage);

//The broadcast outputs the image data in the form of 0 bitmap array
1 save the image output path
intentImage.putExtra("outputImageMode", 0);

//Set the quality percentage of the returned image: 1-100, the
system defaults to 50(50%)
intentImage.putExtra("jpegQuality", 100);

//Whether to return the coordinate position
intentImage.putExtra("outputBounds", true);

//Save the image suffix type 0 png 1 jpg 2 bmp
intentImage.putExtra("imageFormat", 0);

//Set the image save path
intentImage.putExtra("saveImageDirPath", "/storage/emulated/0/imageSa
ve/");
context.sendBroadcast(intentImage);
} catch (Exception e) {
e.printStackTrace();
}
}

```

#### 4.2.2 Registering for image reception broadcast

```

IntentFilter filter = new IntentFilter();
filter.addAction(BROADCAST_IMAGE_ACTION);

```

```
registerReceiver(mGetBitmapReceiver, filter);
```

#### 4.2.3 Listening for image broadcast callbacks

```
BroadcastReceiver mGetBitmapReceiver = new BroadcastReceiver() {

    @Override
    public void onReceive(Context context, Intent intent) {
        String action = intent.getAction();
        if (intent.getAction().equals(BROADCAST_IMAGE_ACTION)) {
            //Get the scan result
            String barcode = intent.getStringExtra("barcode");
            //Get the coordinate set
            int[] bounds = intent.getIntArrayExtra("bounds");
            //Get the image name, for example: 123456787654.jpg
            String imageName = intent.getStringExtra("imageName");
            //Get the image array
            byte[] bmp = intent.getByteArrayExtra("bitmapBytes");
            if (bmp != null && bmp.length > 0) {
                Bitmap mBitmap = BitmapFactory.decodeByteArray(bmp, 0,
                    bmp.length).copy(Bitmap.Config.RGB_565, true);
            }
        }
    }
};
```

#### 4.2.4 Cancellation of registration

If you no longer want to receive pictures, you can cancel your registration.

```
unregisterReceiver(mGetBitmapReceiver);
```

## 5 Broadcast settings scan parameters ( supported by Android 11 and above )

**Broadcast the ACTION for setting the scan code parameters**

|        |                                         |
|--------|-----------------------------------------|
| action | com.android.broadcast.uscanner.settings |
|--------|-----------------------------------------|

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
```

### 5.1 Scan Engine Switch

|       |                       |
|-------|-----------------------|
| extra | keyInt                |
| Value | int type : -10        |
| extra | valueInt              |
| Value | int type 0: off 1: on |

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
```

```
intent.putExtra("keyInt", -10 );
```

```
intent.putExtra("valueInt", 1 );
```

```
sendBroadcast(intent);
```

### 5.2 Start or stop scanning

|       |                                                 |
|-------|-------------------------------------------------|
| extra | keyInt                                          |
| Value | int type: -11                                   |
| extra | valueInt                                        |
| Value | int type 0: turn off decoding 1: start decoding |

Example:

```

Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", -11 );
intent.putExtra("valueInt", 1);
sendBroadcast(intent);

```

## 5.3 Set the output mode

|       |                                                                                    |
|-------|------------------------------------------------------------------------------------|
| extra | keyInt                                                                             |
| Value | int type: -12                                                                      |
| extra | valueInt                                                                           |
| Value | int type 0: open broadcast mode 1: keyboard mode 2: broadcast mode + keyboard mode |

Example:

```

Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", -12 );
intent.putExtra("valueInt", 0 );
sendBroadcast(intent);

```

## 5.4 Trigger Mode

|       |                                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extra | keyInt                                                                                                                                                                                                              |
| Value | int type: -13                                                                                                                                                                                                       |
| extra | valueInt                                                                                                                                                                                                            |
| Value | int type 0: automatic mode (start when the button is pressed, stop after timeout) 1: continuous scanning mode 2: manual mode (start when the button is pressed, stop when the button is released or timeout occurs) |

Example:

```

Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", -13 );
intent.putExtra("valueInt", 0);
sendBroadcast(intent);

```

## 5.5 Lock Scan

|       |                                                                                   |
|-------|-----------------------------------------------------------------------------------|
| extra | keyInt                                                                            |
| Value | int type: -14                                                                     |
| extra | valueInt                                                                          |
| Value | int type 0: locked, cannot scan after locked 1: unlocked, can scan after unlocked |

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", -14 );
intent.putExtra("valueInt", 1 );
sendBroadcast(intent);
```

## 5.6 Barcode Broadcast Settings

### 5.6.1 Barcode Broadcast Name

|       |                                                        |
|-------|--------------------------------------------------------|
| extra | keyInt                                                 |
| Value | int type: 200000 (PropertyID.WEDGE_INTENT_ACTION_NAME) |
| extra | valueStr                                               |
| Value | String Default is: "android.intent.ACTION_DECODE_DATA" |

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", 200000);
intent.putExtra("valueStr","android.intent.ACTION_DECODE_DATA");
sendBroadcast(intent);
```

### 5.6.2 Barcode key name

|       |                                                            |
|-------|------------------------------------------------------------|
| extra | keyInt                                                     |
| Value | int type: 200002 (PropertyID.WEDGE_INTENT_DATA_STRING_TAG) |
| extra | valueStr                                                   |
| Value | The default value for String type is: "barcode_string"     |

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", 200002 );
intent.putExtra("valueStr","barcode_string");
sendBroadcast(intent);
```



## 5.7 Sound and vibration settings in barcode broadcast mode

### 5.7.1 Sound Settings

|       |                                                     |
|-------|-----------------------------------------------------|
| extra | keyInt                                              |
| Value | int type: 6 (PropertyID.SEND_GOOD_READ_BEEP_ENABLE) |
| extra | valueInt                                            |
| Value | int type 0: no beep 1: short 2: sharp               |

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", 6 );
intent.putExtra("valueInt", 1 );
sendBroadcast(intent);
```

### 5.7.2 Vibration settings

|       |                                                        |
|-------|--------------------------------------------------------|
| extra | keyInt                                                 |
| Value | int type: 7 (PropertyID.SEND_GOOD_READ_VIBRATE_ENABLE) |
| extra | valueInt                                               |
| Value | int type 0: turn off vibration 1: turn on vibration    |

Example:

```
Intent intent = new Intent("com.android.broadcast.uscanner.settings");
intent.putExtra("keyInt", 7 );
intent.putExtra("valueInt", 1 );
sendBroadcast(intent);
```

Note: For other parameters, please refer to the PropertyID class or contact our technical support.